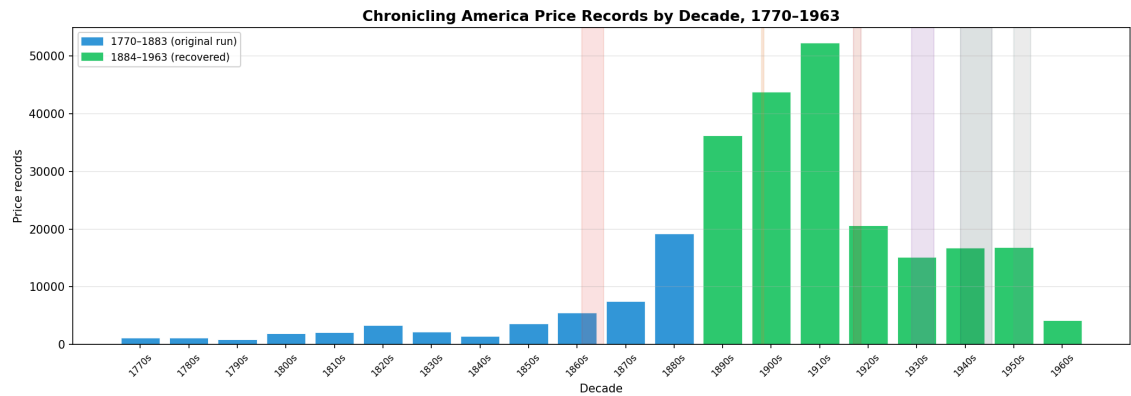


Chronicling America Price Dataset

255,518 structured price records extracted from 112 American newspapers, 1770-1963.

A machine-readable dataset of historical commodity prices, wages, and service costs derived from the Library of Congress [Chronicling America](#) digital newspaper archive using an LLM-based extraction pipeline.



Why This Dataset Exists

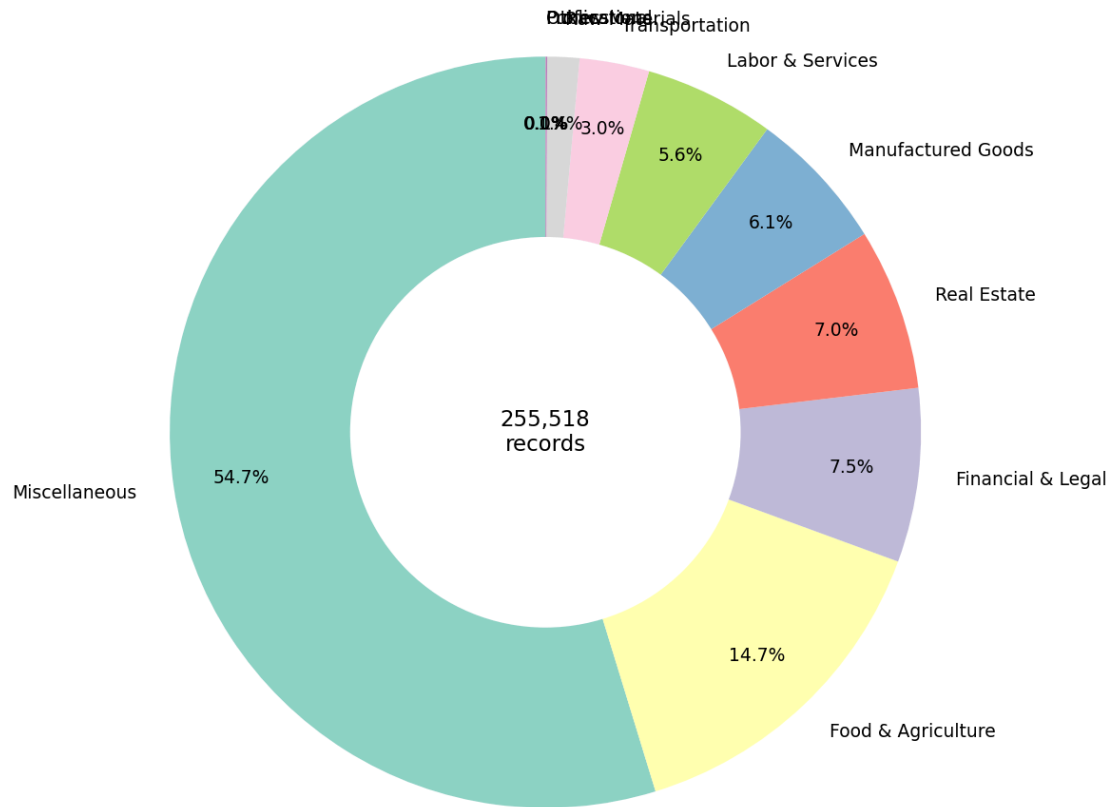
Historical price data before the 20th century is sparse, inconsistent, and trapped in unstructured sources. Government price indices (BLS, NBER) begin reliably only around 1890. Newspaper advertisements and market reports contain rich price information going back to the colonial era - but extracting it at scale requires reading millions of pages of noisy OCR text.

This project applies modern LLMs to that problem: reading OCR text, denoising it, identifying price mentions, classifying commodities, standardizing units, and converting currencies - producing a clean JSONL dataset ready for quantitative analysis.

Dataset Summary

Metric	Value
Total records	255,518
With USD prices	244,551 (95%)
Time span	1770-1963
Newspaper sources	112
Commodity types	79
Economic categories	11

Price Records by Economic Category (full 1770-1963 dataset)



Categories: Food & grain, dairy, meat, sugar, beverages; labor (farm, domestic, professional, unskilled); real estate; transportation; textiles; raw materials; financial instruments; miscellaneous.

Record format (JSONL):

```
{
  "commodity_id": "FOOD-GRAIN-WHEAT",
  "category_ll": "Food & Agriculture",
  "price_per_unit_usd": 1.25,
  "unit": "bushel",
  "year": 1862,
  "location": "Chicago, IL",
  "ref": "sn84026749/1862-03-15/ed-1/seq-4",
  "confidence": 0.92,
  "original_text": "Wheat No.1 Spring $1.25 per bushel"
}
```

Repository Structure

```
data/
  raw/           # Original OCR text from LOC API (by year/newspaper)
  compressed/    # LLM-denoised and compressed OCR (pass 0)
  extracted/     # Structured price mentions extracted by LLM (pass 1)
  normalized/    # Final normalized dataset (pass 2)
    normalized.jsonl  # Main dataset (255,518 records)
    unresolved.jsonl # Records needing manual review
    failed.jsonl     # Processing failures

code/
  download_chronicling_america.py # Step 0: fetch raw OCR from LOC
  compress_pass0.py               # Step 1: LLM denoising/compression
  extract_pass1.py               # Step 2: LLM price extraction
  normalize_prices_pass2.py      # Step 3: LLM normalization + QC
  analyze_prices.py              # Analysis utilities
  generate_report_report_figures.py # Figure generation

report_figures/      # All visualizations
GUIDE.md             # How to use this dataset
```

Pipeline

The extraction pipeline has four stages, each using LLM calls with structured output:

Stage 0 - Download: Fetch OCR text pages from the LOC Chronicling America API for all available newspapers in the target year range.

Stage 1 - Compress: LLM reads raw OCR (often garbled), identifies price-relevant passages, and produces clean compressed text. Reduces data volume ~20x while preserving all price information.

Stage 2 - Extract: LLM identifies individual price mentions in compressed text, outputting structured records with commodity, price, unit, date, and location.

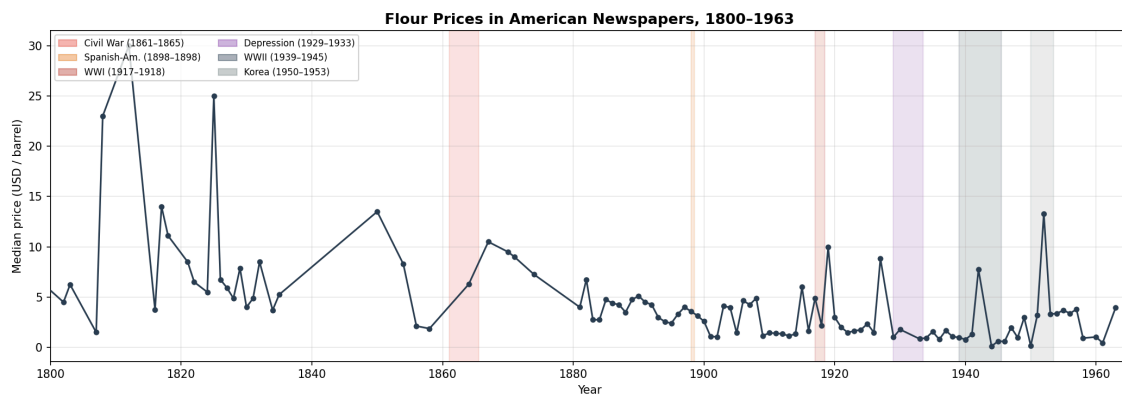
Stage 3 - Normalize: LLM classifies each extracted mention into a 79-commodity taxonomy, standardizes units (bushels, pounds, barrels, etc.), converts historical currencies to USD, and assigns confidence scores. Statistical QC filters apply physical price bounds and outlier removal.

Each stage supports parallelization (`--workers N`) and checkpoint/resume via progress files.

Demo Analysis: What Can You Do With This Data?

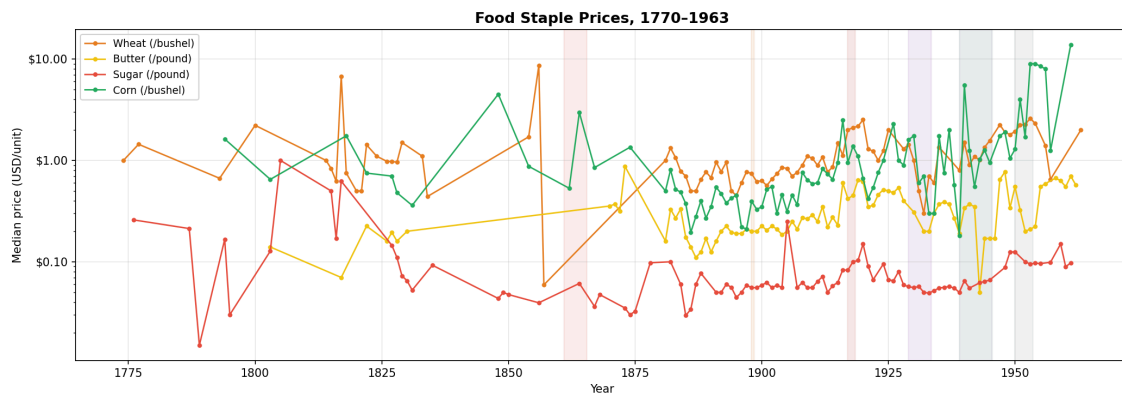
The following analysis demonstrates the dataset's utility for economic history research. Eight hypotheses were tested quantitatively:

Food prices track every American war



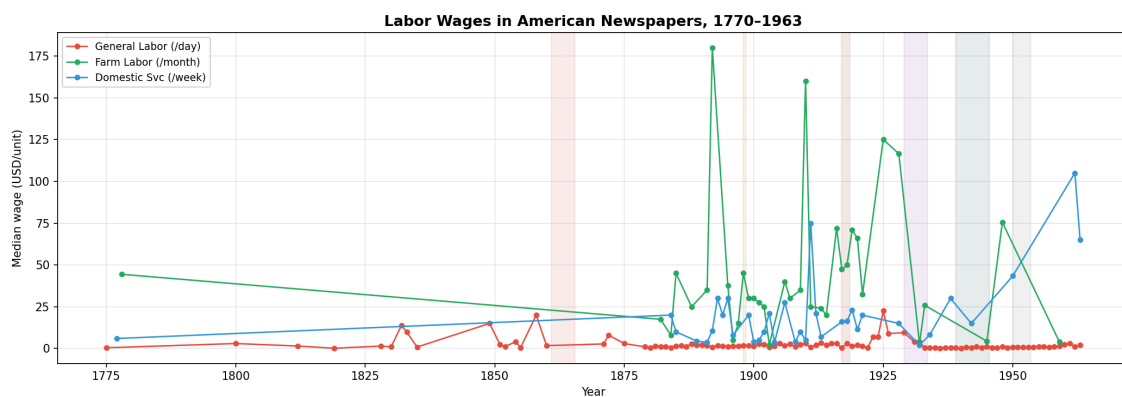
Civil War flour: **+196%**. WWI wheat: **+132%**, flour: **+263%**. WWII (with OPA controls): only +25% — proving price controls worked, though inflation surged +96% immediately after lifting them.

The Great Depression collapsed agricultural prices



Wheat fell **-59%**, butter **-58%**, pork **-43%** between 1925-29 and 1930-34.

Wages grew faster than food prices

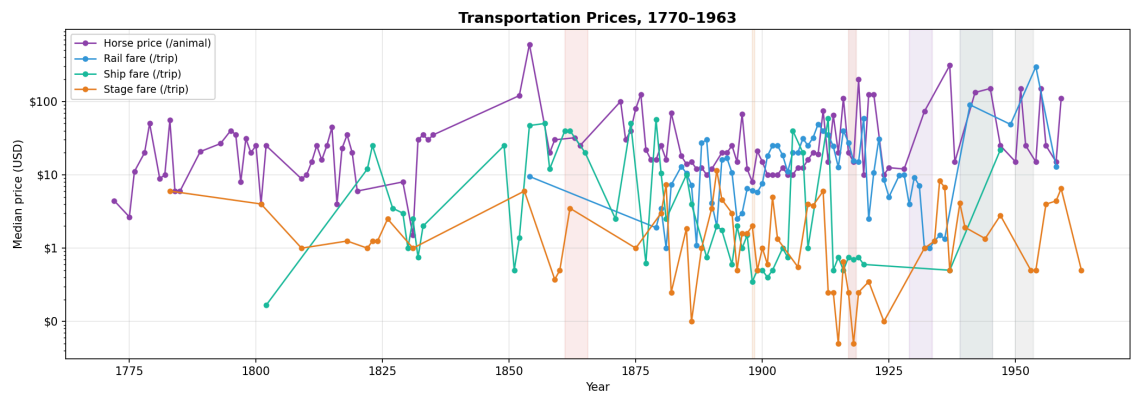


Professional wages: **+76%** (1880-1900 vs 1940-1960). Farm labor: **+73%**. Real purchasing power increased substantially.

The Gold Standard stabilized prices

Wheat price CV during gold standard (1879-1914): **0.254**. After: **0.466**. Post-gold era was **1.8x more volatile**.

Railroads integrated markets

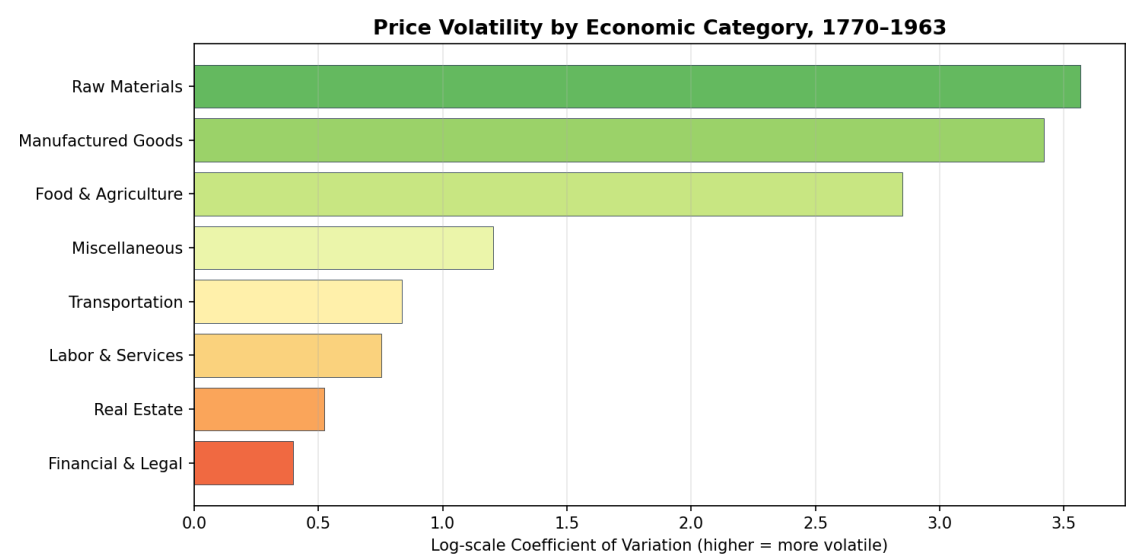


Wheat price dispersion fell **41%** from the pre-railroad era (1820-50) to the mature network (1890-1910).

Agricultural expansion deflated grain prices

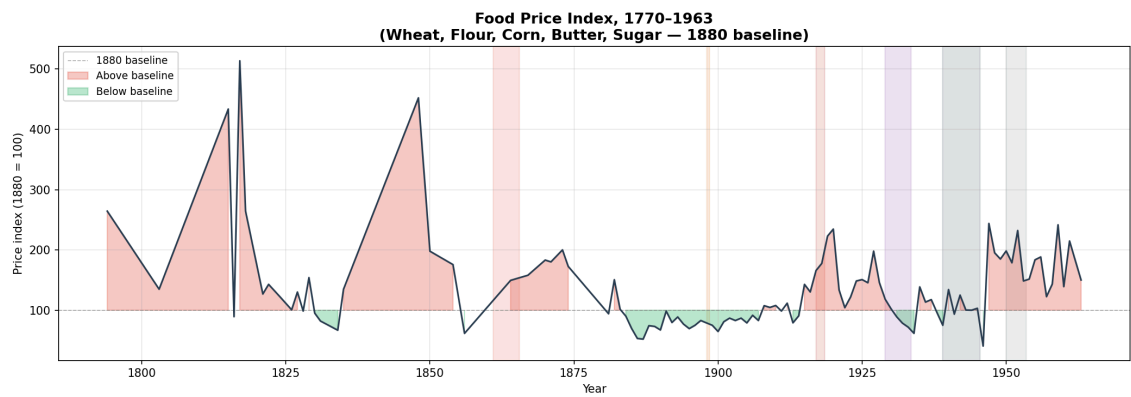
Wheat dropped **38%** and corn **32%** between 1880-85 and 1895-1900 as homesteading opened the Great Plains.

Volatility varies by economic category



Raw materials are most volatile; wages are stickiest.

Composite price index reveals the full arc



Summary of Findings

Hypothesis	Result	Magnitude
Civil War inflation	Confirmed	+196% flour
WWI food inflation	Confirmed	+132% wheat, +263% flour
Great Depression deflation	Confirmed	-59% wheat
Long-run wage growth	Confirmed	+76% professional
Gold Standard stability	Confirmed	1.8x less volatile
Agricultural expansion	Confirmed	-38% wheat
WWII price controls	Confirmed	+25% (vs +132% WWI)
Railroad integration	Confirmed	41% variance reduction

License and Citation

Data derived from the Library of Congress Chronicling America collection (public domain). Processing code and derived dataset released under MIT License.

If you use this dataset, please cite:

Chronicling America Price Dataset, 2024.
255,518 price records from 112 newspapers, 1770-1963.
Source: Library of Congress Chronicling America.

Getting Started

See [GUIDE.md](#) for detailed instructions on using the dataset and running the pipeline.